Lab program no 3:

1) Write a C program to simulate Real-Time CPU Scheduling algorithms:
a) Rate- Monotonic

```
#include <stdio.h>
#include <math.h>
int gcd(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}

int lcm(int a, int b) {
    return (a * b) / gcd(a, b);
}

int main() {
    int n;

    printf("Enter the number of processes:");
    scanf("%d", &n);

    int burst[n], period[n];

    printf("Enter the CPU burst times:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &burst[i]);
    }

    printf("Enter the time periods:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &period[i]);
    }

    int hyper = period[0];
    for (int i = 1; i < n; i++) {
        hyper = lcm(hyper, period[i]);
    }

    printf("LCM=%d\n\n", hyper);
```

```
printf("Rate Monotone Scheduling:\n");
printf("PID\tBurst\tPeriod\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\n", i + 1, burst[i], period[i]);
}
double utilization = 0;
for (int i = 0; i < n; i++) {
    utilization += (double)burst[i] / period[i];
}

double bound = n * (pow(2, (double)1/n) - 1);
printf("\n%f <= %f => %s\n",
       utilization, bound,
       (utilization <= bound) ? "true" : "false");
printf("Scheduling occurs for %d ms\n\n", hyper);
int remaining[n];
for (int i = 0; i < n; i++) {
    remaining[i] = 0;
}
int prev = -2;

for (int time = 0; time < hyper; time++) {
    for (int i = 0; i < n; i++) {
        if (time % period[i] == 0) {
            remaining[i] = burst[i];
        }
    }

    int selected = -1;
    for (int i = 0; i < n; i++) {
        if (remaining[i] > 0) {
            if (selected == -1 || period[i] < period[selected]) {
                selected = i;
            }
        }
    }
    if (selected != prev) {
        if (selected == -1)
            printf("%dms onwards: CPU is idle\n", time);
        else
            printf("%dms onwards: Process %d running\n", time, selected + 1);

        prev = selected;
    }
```

```
        if (selected != -1) {
            remaining[selected]--;
        }
    }

    return 0;
}
```

```
Enter the number of processes:2
Enter the CPU burst times:
20 35
Enter the time periods:
50 100
LCM=100

Rate Monotone Scheduling:
PID     Burst   Period
1       20      50
2       35      100

0.750000 <= 0.828427 => true
Scheduling occurs for 100 ms

0ms onwards: Process 1 running
20ms onwards: Process 2 running
50ms onwards: Process 1 running
70ms onwards: Process 2 running
75ms onwards: CPU is idle

Process returned 0 (0x0)   execution time : 10.631 s
Press any key to continue.
```

b) Earliest-deadline

```
#include <stdio.h>
#include <limits.h>

int main() {
    int n = 3;
    int pid[] = {2, 1, 3};
    int burst[] = {2, 3, 2};
    int deadline[] = {4, 7, 8};
    int period[] = {5, 20, 10};

    int remaining[3];

    for (int i = 0; i < n; i++) {
        remaining[i] = burst[i];
    }


    printf("PID\tBurst\tDeadline\tPeriod\n");
    for (int i = 0; i < n; i++) {
        printf("%d\t%d\t%d\t\t%d\n", pid[i], burst[i], deadline[i], period[i]);
    }

    int time = 0;
    int total_time = 20;

    printf("\nScheduling occurs for %d ms\n\n", total_time);

    while (time < total_time) {
        int selected = -1;
        int min_deadline = INT_MAX;

        for (int i = 0; i < n; i++) {
            if (remaining[i] > 0) {
                if (deadline[i] < min_deadline) {
                    min_deadline = deadline[i];
                    selected = i;
                }
            }
        }

        if (selected == -1) {
            printf("%dms : CPU is idle.\n", time);
```

```
        } else {
            printf("%dms : Task %d is running.\n", time, pid[selected]);

            remaining[selected]--;


            if (remaining[selected] == 0) {
                remaining[selected] = burst[selected];
                deadline[selected] += period[selected];
            }
        }

        time++;
    }

    return 0;
}
```

```
PID     Burst   Deadline        Period
2       2       4               5
1       3       7               20
3       2       8               10

Scheduling occurs for 20 ms

0ms : Task 2 is running.
1ms : Task 2 is running.
2ms : Task 1 is running.
3ms : Task 1 is running.
4ms : Task 1 is running.
5ms : Task 3 is running.
6ms : Task 3 is running.
7ms : Task 2 is running.
8ms : Task 2 is running.
9ms : Task 2 is running.
10ms : Task 2 is running.
11ms : Task 3 is running.
12ms : Task 3 is running.
13ms : Task 2 is running.
14ms : Task 2 is running.
15ms : Task 2 is running.
16ms : Task 2 is running.
17ms : Task 1 is running.
18ms : Task 1 is running.
19ms : Task 1 is running.

Process returned 0 (0x0)   execution time : 0.007 s
Press any key to continue.
```

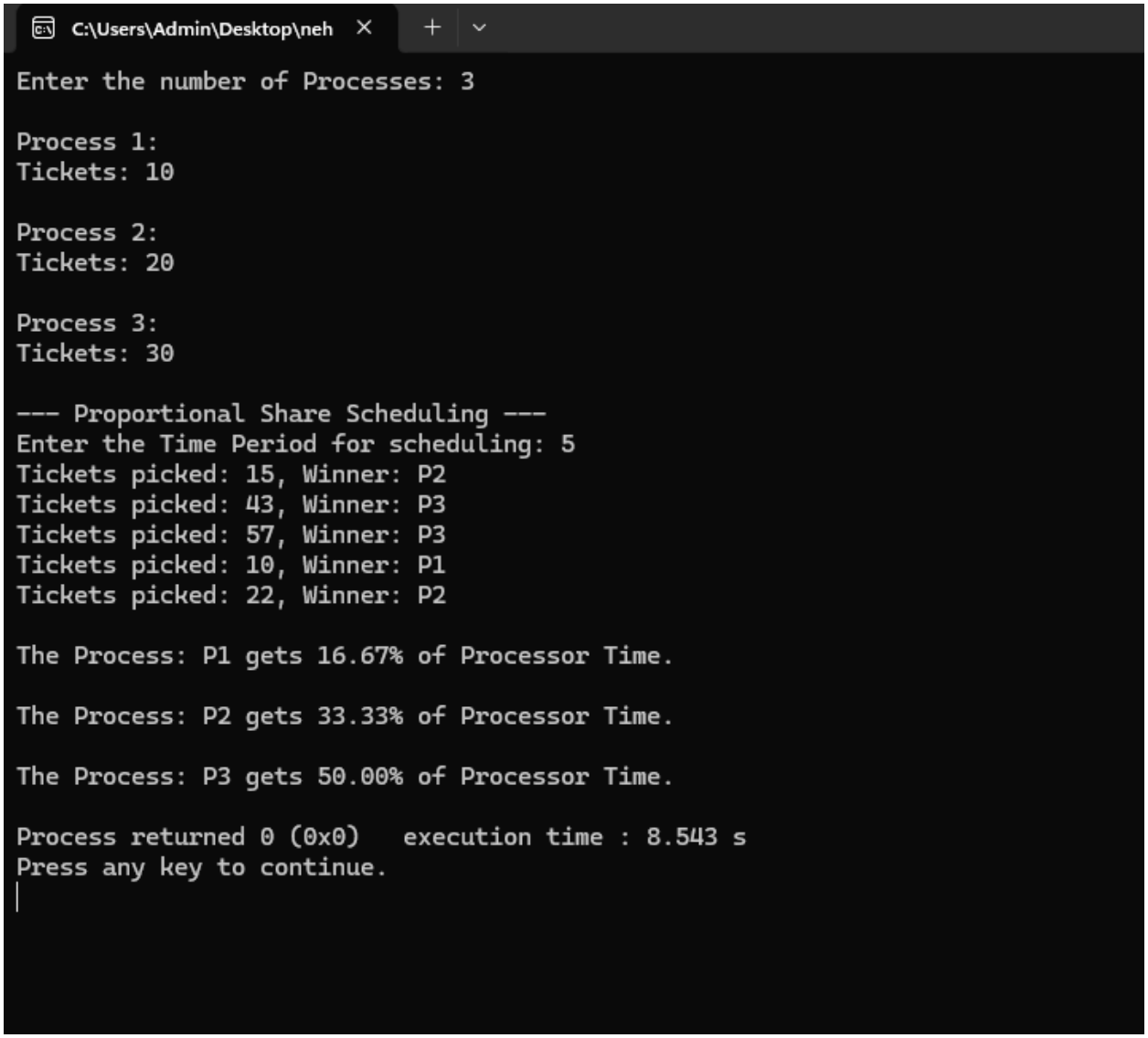
c)Proportional share scheduling

```
#include<stdio.h>
#include <stdlib.h>
#include <time.h>
typedef struct
{
char name[5]; int tickets; }
Process;
```

```
int main() {
int n, total_tickets = 0; float total_T =0.0;
printf("Enter the number of Processes: ");
scanf("%d", &n);
Process p[n];
srand(time(NULL));
for (int i = 0; i < n; i++)
{ printf("\nProcess %d:\n", i + 1);
sprintf(p[i].name, "P%d", i + 1);
printf("Tickets: ");
scanf("%d", &p[i].tickets);
total_tickets += p[i].tickets;
total_T +=p[i].tickets;
}
printf("\n--- Proportional Share Scheduling ---\n");
printf("Enter the Time Period for scheduling: ");
int m;

scanf("%d",&m);
for (int i = 0; i < m; i++)
{
int winning_ticket = rand() % total_tickets + 1;
int accumulated_tickets = 0;
int winner_index;
for (int j = 0; j < n; j++)
{
accumulated_tickets += p[j].tickets;
if (winning_ticket <= accumulated_tickets)
{
winner_index = j; break;
}
}
printf("Tickets picked: %d, Winner: %s\n", winning_ticket,
p[winner_index].name);
}
for (int i = 0; i < n; i++)
{
printf("\nThe Process: %s gets %0.2f%% of Processor Time.\n", p[i].name,
((p[i].tickets / total_T) * 100));

} return 0;
}
```


C:\Users\Admin\Desktop\neh
Enter the number of Processes: 3
Process 1:
Tickets: 10
Process 2:
Tickets: 20
Process 3:
Tickets: 30
--- Proportional Share Scheduling ---
Enter the Time Period for scheduling: 5
Tickets picked: 15, Winner: P2
Tickets picked: 43, Winner: P3
Tickets picked: 57, Winner: P3
Tickets picked: 10, Winner: P1
Tickets picked: 22, Winner: P2
The Process: P1 gets 16.67% of Processor Time.
The Process: P2 gets 33.33% of Processor Time.
The Process: P3 gets 50.00% of Processor Time.
Process returned 0 (0x0) execution time : 8.543 s
Press any key to continue.